

TCS NQT: Simplification & Approximation Masterclass

From the desk of your Senior Examiner (10+ years experience).

1. The Examiner's Cheat Sheet

BODMAS Mastery

V - B - O - D - M - A - S

1. **V**inculum (Bar above numbers, e.g., $\overline{5-3}$)
2. **B**rackets: () then {} then []
3. **O**f (Multiplication, but takes precedence over normal multiplication)
4. **D**ivision & **M**ultiplication (Left to right)
5. **A**ddition & **S**ubtraction (Left to right)

Approximation Tricks

- **Rounding Rules:** If a decimal is .5 or greater, round up; otherwise, round down. Be consistent.
- **Percentage Conversions:** Memorize $1/8 = 12.5\%$, $1/6 = 16.66\%$, $1/9 = 11.11\%$.
- **Smart Rounding:** If multiplying \$98 times 4.9\$, use \$100 times 5 = 500\$ and subtract the error.

2. Examiner's Trap Zone

Trap 1: The "Of" Hierarchy

Many students treat "Of" as simple multiplication and do it after Division. "Of" has higher precedence. $10 \div 2 \text{ of } 5 = 10 \div 10 = 1$, NOT $(10 \div 2) \times 5 = 25$.

Trap 2: Premature Rounding

Rounding every single step in a multi-step calculation leads to "drift" error. Round only at the *end* or use wider bounds if the options are far apart.

3. 10 High-Priority Questions

Q1: $(444 \div 8) \times (24 \div 3) + 120 = ?$

Sol: $55.5 \times 8 + 120 = 444 + 120 = 564.$

Q2: $45.02\% \text{ of } 300 + \sqrt{960} \times 2.1 = ?$

Sol: Approx 45% of 300 + $\sqrt{961} \times 2 = 135 + 31 \times 2 = 135 + 62 = 197.$

Q3: $15/5 + 3(4 - 2) = ?$

Sol: $3 + 3(2) = 3 + 6 = 9.$

Q4: Evaluate $(2x + 3y)^2 - (2x - 3y)^2$.

Sol: Using $a^2 - b^2 = (a+b)(a-b)$: $[(2x+3y) + (2x-3y)] * [(2x+3y) - (2x-3y)] = (4x)(6y) = 24xy.$

Q5: $794.89 \div 15.001 + 767.99 \div 12.001 = ?$

Sol: $795/15 + 768/12 = 53 + 64 = 117.$

Q6: $39.012 \times 14.98 - 28.013 \times 9.999 = ?$

Sol: $39 \times 15 - 28 \times 10 = 585 - 280 = 305.$

Q7: Simplify: $0.2\overline{\text{bar}}(15) + 0.3\overline{\text{bar}}(42)$

Sol: $(215-2)/990 + (342-3)/990 = 213/990 + 339/990 = 552/990 = 92/165.$

Q8: Evaluate $\sqrt[3]{0.000512}$

Sol: $(0.000512)^{(1/3)} = (512/1000000)^{(1/3)} = 8/100 = 0.08.$

Q9: Solve $5x + 3 = 28$

Sol: $5x = 25$, $x = 5$.

Q10: $\frac{1}{2}$ of $\frac{2}{3}$ of $\frac{3}{4}$ of $\frac{4}{5}$ of 100

Sol: $\frac{1}{5} * 100 = 20$.